

**SingleRAN**

# **Time Management Feature Parameter Description**

|              |            |
|--------------|------------|
| <b>Issue</b> | Draft A    |
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# 1 Change History

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This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes  
Changes in functions and their corresponding parameters
- Editorial changes  
Improvements or revisions to the documentation

## 1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-07-30).

### Technical Changes

None

### Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

---

This document describes the Time Management feature, including its basic principles, related features, network impact, and engineering guidelines.

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

#### NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

Unless otherwise specified, in this document, LTE, eNodeB, and eRAN always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, LTE NB-IoT, LTE FDD eNodeB, LTE TDD eNodeB, LTE NB-IoT eNodeB, LTE FDD eRAN, LTE TDD eRAN, and LTE NB-IoT eRAN are used. The "L", "T", and "M" in RAT acronyms refer to LTE FDD, LTE TDD, and LTE NB-IoT, respectively.

In this document, NR and gNodeB always include FDD and TDD.



# 3 Introduction

---

All base stations in a network must have synchronized clocks to operate properly. Failure to keep time synchronized results in the following impacts on network O&M:

- **Inaccuracy of performance statistics:** Inconsistency of the performance statistics collection time occurs. This may lead to confusion in performance reports and traffic-related analysis.
- **Wrong or missing triggering of scheduled tasks:** For example, a license may take effect ahead of schedule.
- **Confusion in problem location:** When the NE reports alarms, the time is inconsistent with the log time. This may cause confusion in problem location and dependency analysis.
- **Base station link setup failure:** If time is not synchronized among the base station, OSS, certificate server, core network, neighboring base station and FTP server, then base station deployment, OMCH setup, IPsec channel setup, DTLS setup, and FTPS setup may fail.

Currently, base stations use the Coordinated Universal Time (UTC).

## Time Zone

A time zone is a geographical region in which the same standard time is observed. The globe is divided into 24 time zones, each of which is, on average, 15° of longitude. The time difference between two neighboring time zones is one hour.

The time of zero-time zone can be represented by Greenwich Mean Time (GMT) or Coordinated Universal Time (UTC). The time of other time zones is represented in relation to zero-time zone. For example, Beijing is located in GMT+8, and the Beijing time can be expressed as GMT+08:00 or UTC+08:00.

## Daylight Saving Time

Many countries, and sometimes certain regions of countries, adopt daylight saving time (DST) during part of the year in compliance with laws and regulations to compensate for the seasonal difference in sunlight and save energy. DST involves advancing clocks by an hour before the start of summer and adjusting the time back by an hour in autumn.

# 4 Time Management

---

## 4.1 Principles

Time sources used by the base station during the normal operation include the Network Time Protocol (NTP) server, Simple Network Time Protocol (SNTP) server, global navigation satellite system (GNSS), and system clock (SYSCLK for short). Base stations of different radio access technologies (RATs) synchronize internal clocks with different time sources:

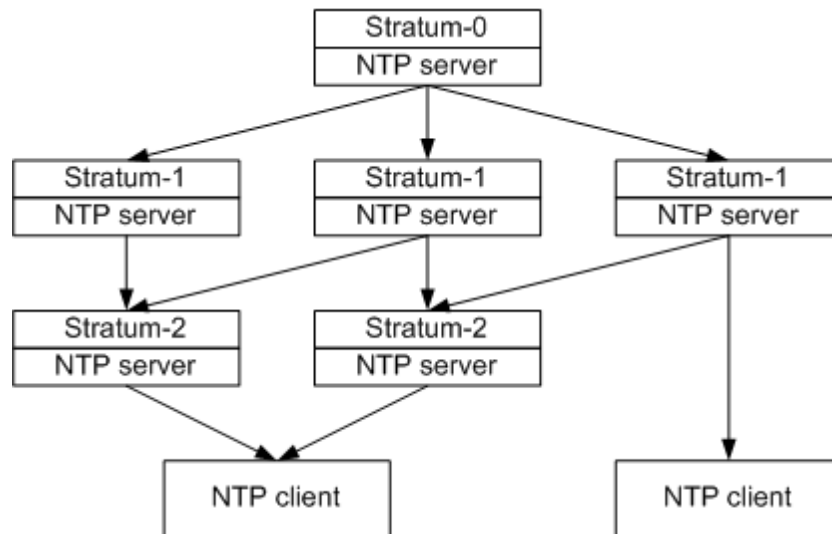
- A GBTS synchronizes its clock with that of a BSC every 30 seconds. The BSC uses an SNTP server as the time source.
- An eGBTS, NodeB, eNodeB, or gNodeB uses an NTP server, GNSS, or SYSCLK as the time source.

### 4.1.1 NTP Server

NTP enables a computer to synchronize with an NTP server or other time sources (such as a quartz clock or GNSS clock) over a wide area network (WAN) or local area network (LAN). NTP time synchronization has a small margin of error (less than 1 ms over a LAN or tens of milliseconds over a WAN) and provides a ciphertext-based authentication mechanism to prevent malicious protocol attacks.

NTP time synchronization is mature and can be implemented as long as the network from a device or lower-stratum server to an upper-stratum server is reachable. The NTP time synchronization period is configurable. [Figure 4-1](#) shows the networking for NTP time synchronization.

**Figure 4-1** Networking for NTP time synchronization



- Top NTP server (stratum-0 NTP server): This server provides time synchronization services for lower-stratum devices.
- Intermediate NTP servers (stratum-1 and stratum-2 NTP servers): These servers synchronize their clocks with the upper-stratum NTP server(s) and provide time synchronization services for lower-stratum devices.
- NTP client: The NTP client synchronizes its clock with one or more intermediate NTP servers. It does not provide time synchronization services.

**NOTE**

The concepts of the server and the client are relative. The device that provides the time synchronization service is called the server, and the device that receives the time synchronization service is called the client.

If there are NTP servers connected to a base station on the network, you can set NTP servers for the base station. A maximum of four NTP servers can be configured on a base station, with one of them as the primary NTP server. If the base station fails to synchronize time with the primary NTP server, it can synchronize time with another NTP server to ensure time synchronization reliability.

Base stations support NTP time synchronization in plaintext or encrypted mode. The encrypted mode improves time synchronization security. If an operator plans to use the security mode for time synchronization, the encryption algorithm used between the NTP server and the base station can be defined. The encryption algorithm must comply with the NTP RFC1305 protocol.

**NOTE**

Simple Network Time Protocol (SNTP) is a simple implementation of NTP. SNTP uses a simplified time synchronization and adjustment algorithm, and its time synchronization accuracy is lower than that of NTP.

## 4.1.2 GNSS Time Source

If a base station uses the GNSS as the time source, it can select the GNSS time source for time synchronization. The GNSS receive device installed on the base

station side can trace and receive signals from multiple GNSS satellites to obtain UTC-based standard time information. If the GNSS time source is faulty after the base station successfully synchronizes its time with the time source, the time source can still provide time synchronization services for the base station for more than 16 hours.

The base station synchronizes its time with the GNSS time source once every 30 minutes.

In GNSS time synchronization, the base station supports enhanced time source processing. This function is controlled by the **Time Source Enhanced Processing Switch** (**TIMETHRD.TIMESRCENHPROC SW (LTE eNodeB, 5G gNodeB)**), which is turned on by default and can be configured using the **SET TIMETHRD** command. The mechanism of this function is as follows:

The base station uses the GNSS time obtained for the first time after the **Time Source Enhanced Processing Switch** is turned on as the reference time and records the time obtained from the system hardware clock. During each subsequent time synchronization, the base station compares the difference ( $\Delta T_G$ ) between two consecutive GNSS time readings with the difference ( $\Delta T_S$ ) between two consecutive system hardware clock readings.

- If the gap between  $\Delta T_G$  and  $\Delta T_S$  is less than or equal to 10 minutes, the base station operates according to the settings of **Threshold Value** (**TIMETHRD.THRD (LTE eNodeB, 5G gNodeB)**) and **Time Hopping Processing Mode** (**TIMETHRD.PROCMODE (LTE eNodeB, 5G gNodeB)**). The reference time is then updated to the GNSS time obtained at each successful time synchronization.
- If the gap between  $\Delta T_G$  and  $\Delta T_S$  is greater than 10 minutes, GNSS time experiences an unexpected jump, causing this time synchronization to fail. When time synchronization fails, the base station operates according to the setting of **Automatic Switch** (for details, see [4.1.5 Time Source Switchover](#)). If this switch is turned on, the base station automatically switches to another time source for time synchronization. If this switch is turned off, the base station waits for the next GNSS time synchronization.

If the base station fails to synchronize with the GNSS time source for 30 days, it will update the reference time using the first GNSS time obtained after the thirtieth day.

#### NOTE

When the **Time Source Enhanced Processing Switch** is turned on and the GNSS time obtained for the first time is abnormal, the abnormality cannot be identified by the base station. In this case, you are advised to run the **SET TIME** command to manually set the current time and then use another time source for time synchronization. In addition, you are advised to run the **SET TIMETHRD** command to turn off the **Time Source Enhanced Processing Switch** and then locate the cause of the abnormality. After the GNSS time becomes normal, the GNSS time source can be used for time synchronization and the **Time Source Enhanced Processing Switch** can be turned on.

## 4.1.3 SYSCLK Time Source

When **Clock Synchronization Mode** is set to **TIME** and the SYSCLK is available, the base station can obtain the UTC-based standard time information from the SYSCLK. The time information of the SYSCLK is derived from a reference clock

source such as the GNSS or IPCLK. For details about the reference clock sources, see *Synchronization Feature Parameter Description* of the corresponding RAT.

If the reference clock source of the SYSCLK is faulty after the base station successfully synchronizes its time with the SYSCLK, the SYSCLK can still provide time synchronization services for the base station for a period of time. The specific duration depends on the selected reference clock source. If the GNSS clock source is selected, the duration is more than 16 hours. For other clock sources such as the IPCLK, the duration is longer than 6 hours.

The base station synchronizes its time with the SYSCLK time source once every 30 minutes.

When a base station uses SYSCLK for time synchronization and the SYSCLK time information is obtained from an independent GNSS reference clock source, the base station supports enhanced time source processing. For details about the processing mechanism, see [4.1.2 GNSS Time Source](#).

## 4.1.4 Time Synchronization

### Initial Time Synchronization

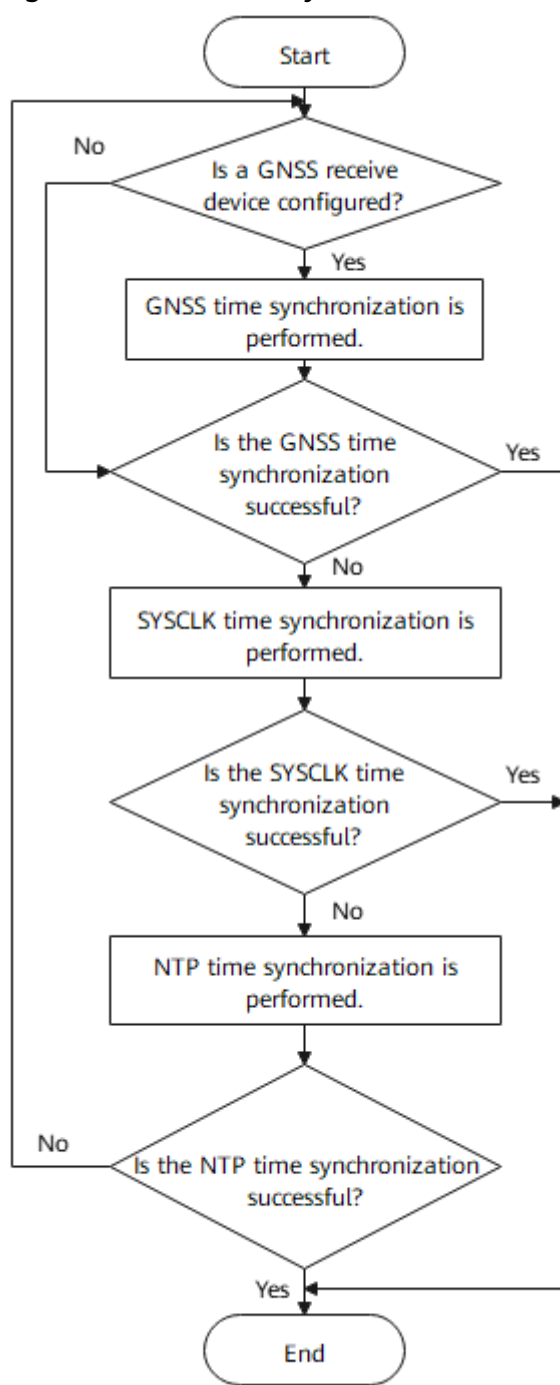
Initial time synchronization is an event that a base station synchronizes its time for the first time when it starts or is reset.

During initial time synchronization, the GBTS proactively sends a time synchronization request to the SNTP server. If time synchronization fails, the GBTS reports ALM-20714 OMU Time Synchronization Abnormity and performs time synchronization again.

During initial time synchronization, the eGBTS, NodeB, eNodeB, or gNodeB sends a time synchronization request to the GNSS receive device, SYSCLK, or NTP server and preferentially selects the time source specified on the base station (by **SET TIMESRC**) for time synchronization. If the time synchronization with the specified time source fails, base station synchronizes its time with other time sources. If the base station successfully synchronizes time with any time source, the initial time synchronization is completed. If the base station fails to synchronize time with all time sources, the base station reports ALM-26266 Time Synchronization Failure and performs time synchronization again.

The following figure shows the initial time synchronization process of the eGBTS, NodeB, eNodeB, and gNodeB using the GNSS time source as an example.

**Figure 4-2** Initial time synchronization



## Periodic Time Synchronization

After initial time synchronization, the base station performs periodic time synchronization based on the time synchronization period of the time source to reduce the time error of the base station relative to the time source and ensure the time accuracy of the base station.

 NOTE

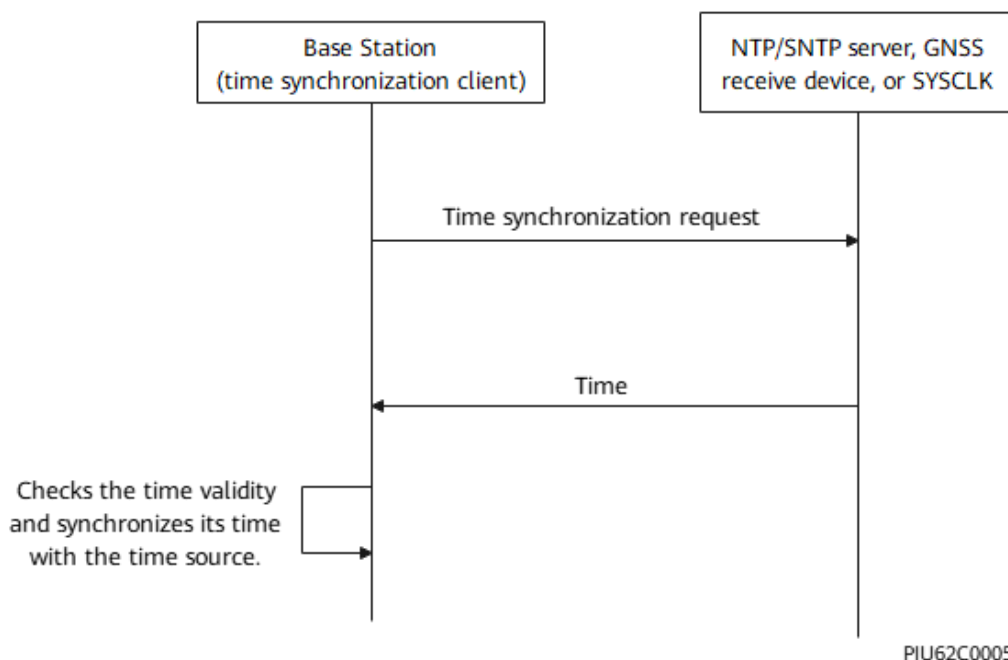
- The **Synchronization Period** of NTP time synchronization is configured by running the **ADD NTPC** or **MOD NTPC** command.
- The **Sync Period** of SNTP time synchronization is configured by running the **SET SNTPCLTPARA** command.
- The synchronization period of GNSS time synchronization and SYSCLK time synchronization is always 30 minutes.

The periodic time synchronization process of the base station is as follows:

1. The base station sends a time synchronization request message to the time source.
2. After receiving the time synchronization request, the time source sends the time to the base station.
3. The base station receives the time and checks its validity. If the time is valid, the base station synchronizes its time with the time source. If the time is invalid, the base station does not synchronize its time (which means the time synchronization fails) and waits for the next synchronization. For NTP time synchronization, if the **Synchronization Period** is set to a value greater than 2 hours, the base station initiates the next time synchronization 2 hours after the periodic NTP time synchronization fails.

The following figure shows the periodic time synchronization process.

**Figure 4-3** Periodic time synchronization



NTP/SNTP time synchronization requires that the routing and VLAN configurations between the base station and the NTP/SNTP server are correct, and the IP transmission link between the base station and the NTP/SNTP server is normal. For details about the prerequisites for GNSS time synchronization and SYSCLK time synchronization, see *Synchronization Feature Parameter Description* of the corresponding RAT.

## 4.1.5 Time Source Switchover

If a base station fails to synchronize its time with the current time source, the base station can automatically switch to another time source for time synchronization, improving the reliability of time synchronization. If time synchronization with another time source is successful, the base station reports EVT-26360 Time Source Switchover. The automatic time source switchover function is controlled by the **Automatic Switch** parameter. If this parameter is set to **ON**, this function is enabled.

### NOTE

You can query the value of the **Time Source** parameter by running the **DSP TIMESRC** command to obtain the time source used by a base station.

Automatic time source switching is classified into the following types:

- Automatic switchover between NTP servers  
Multiple NTP servers can be configured for NTP time synchronization. If the base station fails to synchronize time with the master NTP server, it automatically synchronizes time with other NTP servers.
- Automatic switchover between different types of time sources  
The eGBTS, NodeB, eNodeB, or gNodeB preferentially synchronizes its time with the time source configured in **SET TIMESRC**. If the synchronization fails, it will automatically switch to another type of time source. The automatic time source switchover function applies to the GNSS, SYSCLK, and NTP time sources.

### NOTE

- If the **Automatic Switch** parameter is set to **OFF** and the base station fails to synchronize time with the current time source, the base station will not automatically switch to another time source.
- If an eGBTS, NodeB, eNodeB, or gNodeB is configured with multiple time sources and all the time sources are faulty, the base station time synchronization will fail. The base station will then report ALM-26266 Time Synchronization Failure.

## 4.2 Network Analysis

### 4.2.1 Benefits

N/A

### 4.2.2 Impacts

N/A

## 4.3 Requirements



## 4.3.1 Licenses

None

## 4.3.2 Hardware

### Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

### Boards

A GNSS satellite card is required for GNSS time synchronization.

## 4.3.3 Others

- NTP time synchronization: The transport network required for NTP time synchronization is ready for the base station. If the transport network has security requirements, the NTP server must support encrypted time synchronization.
- SNTP time synchronization: The transport network required for SNTP time synchronization is ready for the base station. If the transport network has security requirements, the SNTP server must support encrypted time synchronization.
- SYSCLK time synchronization: The reference clock source of the SYSCLK is available.

## 4.4 Operation and Maintenance

### 4.4.1 When to Use

Time information is key to functions such as troubleshooting, alarm management, log management, and performance measurement. This section describes how to configure time information for an eGBTS, NodeB, eNodeB, or gNodeB.

#### NOTE

The GBTS obtains time information from the BSC. For details about configuring the time of the BSC, see either of the following sections: **BSC6900 GU Initial Configuration (MML-based) > Configuring the Equipment Data > Configuring the Time** in *BSC6900 GU Initial Configuration Guide* or **BSC6910 GU Initial Configuration (MML-based) > Configuring the Equipment Data > Configuring the Time** in *BSC6910 GU Initial Configuration Guide*.

### 4.4.2 Data Configuration

#### 4.4.2.1 Data Preparation

Data preparation involves two parts:

- Common data, which is collected in all scenarios, such as the time zone, time difference, and daylight saving time (DST) of the base station
- Scenario-specific data, which is collected for specific application scenarios

Collect both required data and scenario-specific data based on requirements.

When the time source is NTP, collect information about the NTP server. By default, the SHA256 authentication mode of NTPv3 is used for NTP time synchronization of the base station. Therefore, the corresponding authentication mode, authentication key, and key index from the NTP server must be obtained. The authentication mode, key, and key index must be consistent between the base station and the NTP server. Otherwise, time synchronization fails.

#### 4.4.2.1.1 Common Data

Before deploying the feature on a base station, determine the time source.

- Configure the time source using the **TIMESRC** MO. [Table 4-1](#) describes the required parameters.

**Table 4-1** Parameters in the **TIMESRC** MO

| Parameter Name                         | Parameter ID                                                                | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time Source                            | <b>TIMESRC.TIMESRC</b><br><i>(LTE eNodeB, 5G gNodeB)</i>                    | Set this parameter based on the actual requirement. <ul style="list-style-type: none"><li>• When time sources are not required during commissioning, this parameter can be set to <b>NONE</b>.</li><li>• When this parameter is set to <b>GNSS</b>, the <b>Clock Working Mode</b> parameter cannot be set to <b>FREE</b>.</li><li>• If the base station is vulnerable to interference from pseudolite signals, it is recommended that this parameter be set to <b>NTP</b> instead of <b>GNSS</b>.</li></ul> |
| Automatic Switch                       | <b>TIMESRC.AUTOSWITCH</b><br><i>(LTE eNodeB, 5G gNodeB)</i>                 | The value <b>ON</b> is recommended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Time Source Enhanced Processing Switch | <b>TIMETHRD.TIMESRCE</b><br><b>NHPROC</b> <i>SW (LTE eNodeB, 5G gNodeB)</i> | Set this parameter to <b>ON</b> if enhanced time source processing is required for the base station.                                                                                                                                                                                                                                                                                                                                                                                                        |

- Configure the NE time using the **OMUCURTIME** MO. [Table 4-2](#) describes the required parameters.

 **NOTE**

A base station can automatically synchronize its time during the initial time synchronization phase. Generally, manual time synchronization is not required. However, the time can also be manually set by running MML commands.

**Table 4-2** Parameters in the **OMUCURTIME** MO

| Parameter Name | Parameter ID                                      | Setting Notes    |
|----------------|---------------------------------------------------|------------------|
| Date           | <b>OMUCURTIME.DATE</b><br>(LTE eNodeB, 5G gNodeB) | Set the NE date. |
| Time           | <b>OMUCURTIME.TIME</b><br>(LTE eNodeB, 5G gNodeB) | Set the NE time. |

- Configure the time zone and DST using the **TZ** MO. [Table 4-3](#) describes the required parameters.

**Table 4-3** Parameters in the **TZ** MO

| Parameter Name      | Parameter ID                               | Setting Notes                                                                                                                                                                                                               |
|---------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time Zone           | <b>TZ.ZONET</b> (LTE eNodeB, 5G gNodeB)    | This parameter specifies the time zone of the NE. Set this parameter based on the actual time zone where the NE is located.                                                                                                 |
| Time Difference     | <b>TZ.TIMEDIFF</b> (LTE eNodeB, 5G gNodeB) | This parameter is configurable when the <b>Time Zone</b> parameter is set to <b>CUSTOM</b> or <b>CUSTOM_UTC</b> .<br>Set this parameter based on the time zone where the base station is located.                           |
| DST Tag Information | <b>TZ.DST</b> (LTE eNodeB, 5G gNodeB)      | This parameter specifies whether DST is enabled based on the operator's requirements. Set this parameter to <b>NO</b> when DST is not required. Set this parameter to <b>YES</b> when DST is required for a period of time. |

| Parameter Name             | Parameter ID                                          | Setting Notes                                                                                                                                                                                                                                                                  |
|----------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DST Start Mode             | <b><i>TZ.SM (LTE eNodeB, 5G gNodeB)</i></b>           | Set these parameters based on the time zone where the base station is located.<br><br>If local laws and regulations specify that DST must be used, set the related parameters based on the laws and regulations.                                                               |
| DST Start Month            | <b><i>TZ.SMONTH (LTE eNodeB, 5G gNodeB)</i></b>       |                                                                                                                                                                                                                                                                                |
| DST Start Day              | <b><i>TZ.SDAY (LTE eNodeB, 5G gNodeB)</i></b>         |                                                                                                                                                                                                                                                                                |
| DST Start Week Sequence    | <b><i>TZ.SWSEQ (LTE eNodeB, 5G gNodeB)</i></b>        |                                                                                                                                                                                                                                                                                |
| DST Start Weekday          | <b><i>TZ.SWEEK (LTE eNodeB, 5G gNodeB)</i></b>        |                                                                                                                                                                                                                                                                                |
| DST Start Time             | <b><i>TZ.ST (LTE eNodeB, 5G gNodeB)</i></b>           |                                                                                                                                                                                                                                                                                |
| DST End Mode               | <b><i>TZ.EM (LTE eNodeB, 5G gNodeB)</i></b>           |                                                                                                                                                                                                                                                                                |
| DST End Month              | <b><i>TZ.EMONTH (LTE eNodeB, 5G gNodeB)</i></b>       |                                                                                                                                                                                                                                                                                |
| DST End Day                | <b><i>TZ.EDAY (LTE eNodeB, 5G gNodeB)</i></b>         |                                                                                                                                                                                                                                                                                |
| DST End Week Sequence      | <b><i>TZ.EWSEQ (LTE eNodeB, 5G gNodeB)</i></b>        |                                                                                                                                                                                                                                                                                |
| DST End Weekday            | <b><i>TZ.EWEEK (LTE eNodeB, 5G gNodeB)</i></b>        |                                                                                                                                                                                                                                                                                |
| DST End Time               | <b><i>TZ.ET (LTE eNodeB, 5G gNodeB)</i></b>           |                                                                                                                                                                                                                                                                                |
| DST Offset                 | <b><i>TZ.TO (LTE eNodeB, 5G gNodeB)</i></b>           |                                                                                                                                                                                                                                                                                |
| Support Forcible Execution | <b><i>TZ.FORCEEXECUTE (LTE eNodeB, 5G gNodeB)</i></b> | This parameter indicates whether to forcibly set the time zone and DST based on the operator's requirements. When this parameter is set to <b>NO</b> , the time zone and DST cannot be set. When this parameter is set to <b>YES</b> , the time zone and DST are forcibly set. |

### 4.4.2.1.2 Scenario-specific Data

## NTP Time Synchronization

Configure the NTP server information by using the **NTPCP** MO. [Table 4-4](#) describes the key parameters.

#### NOTE

- If an operation and maintenance channel (OMCH) is configured, the NTP client uses the OMCH local IP address to communicate with the NTP server. If no OMCH is configured, the NTP client automatically uses a device IP address of the base station based on the route information to communicate with the NTP server.
- If an IPv4-based OMCH and an IPv6-based OMCH work in active/standby mode, two NTP clients must be configured using the **NTPCP** MO. The **NTPCP.MODE (LTE eNodeB, 5G gNodeB)** parameter must be set to **IPv4** for one NTP client and to **IPv6** for the other. This prevents NTP time synchronization failures caused by lack of corresponding configurations on the NTP client in the case of the switchover between the IPv4-based and IPv6-based OMCHs.

**Table 4-4** Parameters in the **NTPCP** MO

| Parameter Name             | Parameter ID                                   | Setting Notes                                                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IP Mode                    | <b>NTPCP.MODE (LTE eNodeB, 5G gNodeB)</b>      | Set this parameter to <b>IPv4</b> or <b>IPv6</b> based on the network plan.                                                                                                                                                                                                                                                              |
| IPv4 Address of NTP Server | <b>NTPCP.IP (LTE eNodeB, 5G gNodeB)</b>        | This parameter specifies the IP address of the NTP server. Set this parameter only when the IP mode of the NTP client is IPv4.                                                                                                                                                                                                           |
| IPv6 Address of NTP Server | <b>NTPCP.IPV6 (LTE eNodeB, 5G gNodeB)</b>      | This parameter specifies the IP address of the NTP server. Set this parameter only when the IP mode of the NTP client is IPv6.                                                                                                                                                                                                           |
| Synchronization Period     | <b>NTPCP.SYNCCYCLE (LTE eNodeB, 5G gNodeB)</b> | This parameter specifies the period based on which NTP time synchronization is performed periodically.                                                                                                                                                                                                                                   |
| Authentication Mode        | <b>NTPCP.AUTHMODE (LTE eNodeB, 5G gNodeB)</b>  | This parameter specifies the NTP authentication mode. The default value is <b>SHA256</b> . Time synchronization requires that the NTP authentication mode on a base station be the same as that configured on the NTP server. Otherwise, time synchronization will fail. The SHA256 authentication mode can improve the system security. |

| Parameter Name           | Parameter ID                                             | Setting Notes                                                                                                                                                                                   |
|--------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Key       | <b>NTPCP.KEY</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        | This parameter specifies the key used for NTP authentication. Time synchronization requires that the NTP authentication key on a base station be the same as that configured on the NTP server. |
| Authentication Key Index | <b>NTPCP.KEYID</b> ( <i>LTE eNodeB, 5G gNodeB</i> )      | This parameter specifies the server-side index of the NTP authentication key. The index must be the same as the setting on the server.                                                          |
| Port Number              | <b>NTPCP.PORT</b> ( <i>LTE eNodeB, 5G gNodeB</i> )       | This parameter specifies the port number of the NTP server for time synchronization.                                                                                                            |
| Master Flag              | <b>NTPCP.MASTERFLAG</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) | This parameter specifies the flag identifying the master NTP server when multiple NTP servers are configured.                                                                                   |

 **NOTE**

- If the base station uses the ciphertext-based authentication mode for time synchronization, the ciphertext-based authentication mode must be enabled on the NTP server.
- The authentication mode, authentication key, and key index used by the base station must have been configured on the NTP server. Then the base station obtains related information from the NTP server to ensure that the key information configured on the base station is consistent with that on the NTP server. Otherwise, time synchronization will fail.
- The value of **Authentication Key** must comply with the parameter setting rules of MML commands. For details, see *MML Command Overview* in *3900 & 5900 Series Base Station Product Documentation*.

## GNSS Time Synchronization

The parameter configuration procedure for GNSS time synchronization is as follows:

1. Add a GNSS time source by setting the related parameters in the **GNSS MO**. For details, see *Synchronization Feature Parameter Description* of the corresponding RAT.
2. Set parameters in the **TASM MO** such as **Clock Working Mode**.

**Table 4-5** Parameters in the **TASM** MO

| Parameter Name        | Parameter ID                                        | Setting Notes                                                                 |
|-----------------------|-----------------------------------------------------|-------------------------------------------------------------------------------|
| Clock Working Mode    | <b>TASM.MODE</b> ( <i>LTE eNodeB, 5G gNodeB</i> )   | Set this parameter to <b>AUTO</b> or <b>MANUAL</b> based on the service plan. |
| Selected Clock Source | <b>TASM.CLKSRC</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) | Set this parameter to <b>GNSS</b> .                                           |
| Clock Source No.      | <b>TASM.SRCNO</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  | Set this parameter to the number of the created clock link.                   |

## SYSCLK Time Synchronization

Parameters related to SYSCLK time synchronization include parameters in the **TASM** MO and parameters related to the reference clock source. The following table describes the parameters in the **TASM** MO. For details about how to configure the reference clock source, see *Synchronization Feature Parameter Description* of the corresponding RAT.

**Table 4-6** Parameters in the **TASM** MO

| Parameter Name             | Parameter ID                                             | Setting Notes                                                                 |
|----------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------|
| Clock Synchronization Mode | <b>TASM.CLKSYNCMODE</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) | Set this parameter to <b>TIME</b> .                                           |
| Clock Working Mode         | <b>TASM.MODE</b> ( <i>LTE eNodeB, 5G gNodeB</i> )        | Set this parameter to <b>AUTO</b> or <b>MANUAL</b> based on the service plan. |
| Selected Clock Source      | <b>TASM.CLKSRC</b> ( <i>LTE eNodeB, 5G gNodeB</i> )      | Set this parameter based on site requirements.                                |

### 4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

 **NOTE**

If the planned time source is unavailable during the initial stage of base station deployment, run the **SET TIME** command to manually set the current time.

## Scenario 1: NTP Time Synchronization

- Step 1** Run the **ADD NTPC** command with related parameters specified to add an NTP client.
- Step 2** Run the **SET MASTERNTPS** command to set parameters of the master NTP server.
- Step 3** Run the **SET TIMESRC** command with **Time Source** set to **NTP**.

----End

 **NOTE**

Multiple NTP servers can be configured for NTP. When the master NTP server is unavailable, a switchover to an available NTP server occurs. Run the **DSP NTPC** command to query the IP address of the NTP server in use. It is recommended that the current time for the base station be manually configured to ensure accuracy.

## Scenario 2: GNSS Time Synchronization

- Step 1** Run the **ADD GNSS** command with related parameters specified to add a GNSS time source.
- Step 2** Run the **SET CLKMODE** command to configure the working mode of the reference clock source.
- Step 3** Run the **SET TIMESRC** command with **Time Source** set to **GNSS**.

----End

## Scenario 3: SYSCCLK Time Synchronization

- Step 1** Run the **SET CLKSYNCMODE** command with **Clock Synchronization Mode** set to **TIME**.
- Step 2** Run the **ADD GNSS**, **ADD IPCLKLINK**, or other MML commands to add the corresponding reference clock source.
- Step 3** Run the **SET CLKMODE** command to configure the working mode of the reference clock source.
- Step 4** Run the **SET TIMESRC** command with **Time Source** set to **SYSCCLK**.

----End

## MML Command Examples

```
//Setting the time zone and DST
SET TZ: ZONET=GMT+0800, DST=YES, SM=DATEWEEK, SMONTH=JAN, SDAY=1, SWEK=TUE,
ST=10&49&00, EM=DATE, EMONTH=FEB, EDAY=26, ET=10&49&00, TO=30, FORCEEXECUTE=YES;
//Setting the current time
SET TIME: DATE=2015&01&01, TIME=12&00&00;
```



## Scenario 1: NTP Time Synchronization Example

```
//Adding an NTP client  
  
ADD NTPC: MODE=IPV4, IP="192.168.88.168", PORT=123, SYNCYCLE=10, AUTHMODE=SHA256,  
KEY="*****", KEYID=1;  
//Setting parameters of the master NTP server  
SET MASTERNTPS: MODE=IPV4, IP="192.168.88.168";  
//Setting the time source to NTP  
SET TIMESRC: TIMESRC=NTP;  
//Querying the latest successful synchronization time of the base station  
LST LATESTSUCCDATE;;
```

## Scenario 2: GNSS Time Synchronization Example

```
//Adding a GNSS time source  
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=4;  
//Setting the working mode of the reference clock source  
SET CLKMODE: MODE=MANUAL, CLKSRC=GNSS, SRCNO=0;  
//Setting the time source to GNSS  
SET TIMESRC: TIMESRC=GNSS;  
//Querying the latest successful synchronization time of the base station  
LST LATESTSUCCDATE;;
```

## Scenario 3: SYSCLK Time Synchronization Example

```
//Setting the clock synchronization mode of the base station  
SET CLKSYNCMODE: CLKSYNCMODE=TIME;  
//Adding an IP clock link  
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=6, CNM=L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,  
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;  
//Setting the working mode of the reference clock source  
SET CLKMODE: MODE=MANUAL, CLKSRC=IPCLK, SYNMODE=OFF;  
//Setting the time source to SYSCLK  
SET TIMESRC: TIMESRC=SYSCLK;  
//Querying the latest successful synchronization time of the base station  
LST LATESTSUCCDATE;;
```

### 4.4.3 Activation Verification

- Step 1** Run the **SET TIME** command to set the date and time to earlier points in time, for example, to **2012-12-12** and **00:00:00** respectively.
- Step 2** When the time synchronization period elapses, run the **DSP TIME** command to query the current time of the base station. If the value of **Time** in the command output is the same as the time of the server, the base station time is accurate.
- Step 3** Run the **DSP TIMESRC** command to query the current time source. If the value of **Time Source** in the command output is the same as the site configuration, the time source is correctly configured.
- Step 4** Run the **LST LATESTSUCCDATE** command to check whether the time synchronization is successful. If the time synchronization is successful, the Time Management feature has been successfully activated.

----End

### 4.4.4 Network Monitoring

The base station supports time hopping detection. This function can be enabled by running the **SET TIMETHRD** command with **Time Hopping Detection Switch** set to **ON** and **Threshold Value** and **Time Hopping Processing Mode** set to appropriate values.

- EVT-26360 Time Source Switchover will be reported when **Time Hopping Processing Mode** is set to **RPT\_EVENT\_AND\_SYNC** and the difference between the time obtained from an external reference time source and the current time of the local NE exceeds the threshold. Then, time hopping occurs, performance data collection and logging may be incorrect.
- ALM-26266 Time Synchronization Failure will be reported when **Time Hopping Processing Mode** is set to **RPT\_ALARM\_AND\_NOT\_SYNC** and the difference between the time obtained from the external reference time source and the current time of the local NE exceeds the threshold. If this alarm is not handled for a long time, performance data collection and logging may be incorrect.

# 5 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 6 Reference Documents

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- *Synchronization Feature Parameter Description* of the corresponding RAT
- *BSC6900 GU Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*
- *BSC6910 GU Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*